The Squiggly Line Theorem: Squigglier Curves Are Closer to the Data, in a Direction Training Cannot Use

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Draft of 11 November 2026.

Abstract

The squiggly line conjecture is a geometric account of why larger networks learn faster: a network with more parameters is a squigglier random curve at initialisation, a squigglier curve passes nearer to the data, and so a larger network starts closer to a fit. We split it into three claims and settle each as far as we can. The geometric claim is a theorem for sines. For a random-phase sine of frequency w and data whose labels lie in its band, the distance from a data point to the nearest point of the curve lies between the vertical error divided by $\sqrt{1+K^2}$, with K the slope of the sine, and $1/w$; its expectation tends to zero as w grows. The learning claim is false for the same family: the expected loss of a random function depends only on its one-point marginals, and these do not depend on the frequency, so the expected loss is the same at every frequency for every regression loss. The claim about parameters fails for width: under He initialisation the expected initial squared error of a ReLU network is the same at every width, and the curves of random networks do not get longer with width. The idea is true in parameter space. For a model that is linear in its parameters, the distance from the initial parameters to the nearest fit is determined by the kernel, equals the size of the residual divided by $\sqrt{c}$ when the features are orthogonal with squared length c, and never grows when parameters are added; a Fourier-series experiment shows the initial loss flat and the distance to a fit falling steeply with the number of terms. The main statements are checked in Lean 4. For neural networks the geometric claim remains a conjecture, which we reduce to a question about the oscillation scale of a random network.

1 Introduction

Networks with far more parameters than training points are easier to train than small ones, and several accounts of this exist: a large random network contains good sparse subnetworks [Frankle and Carbin, 2019, Ramanujan et al., 2020]; a wide network stays close to its linearisation around the initial weights [Jacot et al., 2018, Chizat et al., 2019]; and test risk can fall again past the interpolation threshold [Belkin et al., 2019, Nakkiran et al., 2020]. This paper examines a simpler geometric picture, which we call the *squiggly line conjecture*. Part of it is a theorem, part of it is false, and what is true of the rest lives in a different space from the one the picture is drawn in.

The picture is as follows. Scatter random points on a plane and draw two curves without looking at the points, one smooth and one squiggly. The squiggly curve passes closer to the points on average, because it visits more of the plane (Figure 1). A network computes a function, and the graph of that function is such a curve. The conjecture says that a network with more parameters is a squigglier curve at initialisation, that it therefore starts closer to its data, and that this is why it learns faster. This is a chain of three claims.

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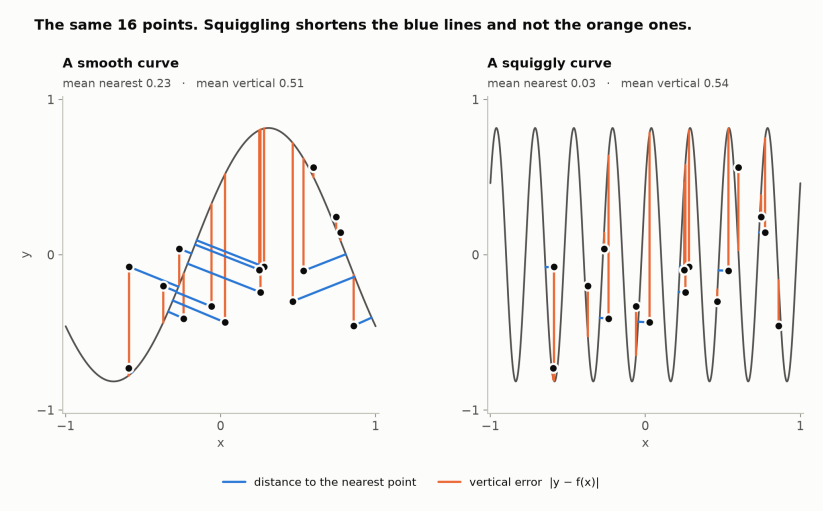


Figure 1: The two distances, on the same 16 points. Blue: distance to the nearest point of the curve. Orange: vertical error $|y - f(x)|$. Raising the frequency of the sine from 1 (left) to 8 (right) shortens the blue segments and leaves the orange ones as long as before.

(G) Geometry. A squigglier curve has a smaller mean Euclidean distance from a data point to the nearest point of the curve.

(P) Parameters. More parameters give a squigglier random function.

(L) Learning. A smaller nearest-point distance means a smaller training loss at initialisation.

Results. (G) is true and is proved for sines, (P) is false for width and holds weakly for depth, and (L) is false. The idea survives in parameter space.

- *The theorem for sines* (Theorem 12). For a random-phase sine of frequency w and labels in its band, the nearest-point distance lies between the vertical error divided by $\sqrt{1+K^2}$ and $1/w$, its expectation tends to zero, and the expected loss is the same at every frequency. The upper bound is an instance of a covering bound (Theorem 11), which reduces (G) for any family to the oscillation scale of its functions.
- *Parameter space* (Section 4). For a model that is linear in its parameters, gradient descent and the distance to the nearest fit depend on the model only through its kernel (Theorem 14); with orthogonal features of squared length c that distance is the size of the residual divided by $\sqrt{c}$ (Corollary 15); and adding parameters never increases it (Proposition 16). In a Fourier-series experiment the initial loss is flat in the number of parameters while the distance to a fit falls by a factor of 2000.
- *Slope sandwich* (Theorem 1). For a K-Lipschitz function and any data point, $\text{vertical}/\sqrt{1+K^2} \le \text{nearest} \le \text{vertical}$. The curve can be much nearer than its vertical error only where it is steep.
- *Marginal lemma* (Theorem 2). The expected loss of a random function, for any loss of the residual $y - f(x)$, depends only on the distribution of $f(x)$ at each single x. For random-phase waves this distribution does not depend on the frequency (Corollary 3), and for ReLU networks under He initialisation the expected initial squared error does not depend on the width (Corollary 4).
- *Tube identity and length bound* (Theorem 5 and Proposition 7). The mean distance from a random point to a set is the mass outside the tube of radius r around the set, integrated over r. For a curve of length l and a uniform point in a planar region of area V it is at least $V/(4l + 2\sqrt{\pi V})$.
- *The sine* (Propositions 8 and 9). For a sine that spans the data, distance times frequency tends to $(2 - 8/\pi^2)/4 \approx 0.2974$. With a smaller amplitude the distance stops at a floor set by the amplitude, not by the number of points.

- *Experiments* (Section 3). With amplitude and marginals held fixed, the nearest-point distance falls from 0.613 to 0.021 while the vertical error stays between 0.665 and 0.669. Random ReLU networks have the predicted, width-independent initial squared error, and their curves are not longer at larger width.

Scope. The theorem is about sines. For neural networks claim (G) remains a conjecture, which Theorem 11 reduces to a question about the oscillation scale of a random network. An earlier draft of this paper claimed empirical support for the conjecture; that support came from faulty measurement code, and the draft was withdrawn (Appendix C). The individual statements here are elementary, and we make no claim of priority for any of them; the parameter-space results in particular are the lazy-training picture in its simplest case. Section 6 lists the work we are aware of. Theorems 1, 2, 5, 11, 12 and 14, Corollaries 3 and 15 and Propositions 13 and 16 are formalised in Lean 4 (Section 5). Code, the Lean project and the development notes are at https://github.com/Ghost---Shadow/squiggly-line-theorem.

2 Setting and results

2.1 Two distances

Let (X, d) be a metric space of inputs and $f : X \to \mathbb{R}$ a function. A data point is a pair $(x, y) \in X \times \mathbb{R}$. We give $X \times \mathbb{R}$ the Euclidean product metric $\rho\big((x, y), (x', y')\big) = \sqrt{d(x, x')^2 + (y - y')^2}$ and write $\Gamma_f = \{(x, f(x)) : x \in X\}$ for the graph of f, the "line" of the conjecture. Two quantities compare a data point with the line:

$$v_f(x, y) = |y - f(x)| \qquad \text{(vertical error)},$$
$$\delta_f(x, y) = \inf_{x' \in X} \rho\big((x, y), (x', f(x'))\big) \qquad \text{(nearest-point distance)}.$$

Regression losses are functions of the residual $y - f(x)$, so they see v_f. The conjecture is stated with δ_f.

A *random function* F is a probability space (Θ, π) of parameters together with functions $f_\theta : X \to \mathbb{R}$ such that $\theta \mapsto f_\theta(x)$ is measurable for every x. Its *one-point marginal* at x is the law ν_x^F of $f_\theta(x)$ under π. For a cost c and a data distribution μ on $X \times \mathbb{R}$, the expected cost at initialisation is $\mathbb{E}_{(x,y)\sim\mu}\, \mathbb{E}_{\theta\sim\pi}\, c(f_\theta; x, y)$.[1] For a family of random functions indexed by a size m, "a bigger model starts closer" means that this expectation is strictly decreasing in m. Claim (G) uses the cost δ and claim (L) uses costs of the form $\ell(y - f(x))$. We make "squiggly" concrete by the frequency of a wave of fixed amplitude and, for networks, by arc length and the number of turns.

2.2 The slope sandwich

Theorem 1 (Slope sandwich)**.** *If f is K-Lipschitz, then for every data point (x, y)*

$$\frac{v_f(x, y)}{\sqrt{1 + K^2}} \;\le\; \delta_f(x, y) \;\le\; v_f(x, y).$$

Proof. The point $(x, f(x))$ lies on Γ_f at distance $|y - f(x)|$ from (x, y), which gives the upper bound. For the lower bound take any $x' \in X$ and put $a = d(x, x')$ and $b = |y - f(x')|$. By the triangle inequality and the Lipschitz bound, $|y - f(x)| \le |y - f(x')| + |f(x') - f(x)| \le b + Ka$. By the Cauchy–Schwarz inequality, $b + Ka \le \sqrt{1 + K^2}\,\sqrt{a^2 + b^2} = \sqrt{1 + K^2}\;\rho\big((x, y), (x', f(x'))\big)$. Taking the infimum over x' gives $v_f(x, y) \le \sqrt{1 + K^2}\,\delta_f(x, y)$. □

Both bounds are attained: for $X = \mathbb{R}$ and $f(x) = Kx$, the point $(0, y)$ has $\delta_f = |y|/\sqrt{1 + K^2}$, and for a constant f the two distances coincide. Taking expectations, the same two inequalities hold between the mean distances of any random K-Lipschitz function and any data distribution.

[1] We assume the inner expectation is a measurable function of (x, y). This holds if $(\theta, x) \mapsto f_\theta(x)$ is jointly measurable, and always when μ is the empirical measure of a finite data set.

The theorem says what a small nearest-point distance means. If $\delta_f(x, y) < \eta$, there is an input x' with $d(x, x') < \eta$ and $|y - f(x')| < \eta$: a perturbation of the *input* smaller than η brings the output within η of the label. When δ_f is small and v_f is not, the function is steep near x: its Lipschitz constant is at least $\sqrt{(v_f/\delta_f)^2 - 1}$. That describes a function that is sensitive to its input near the data, not one that fits the data.

2.3 The marginal lemma

Theorem 2 (Marginal lemma)**.** *Let F be a random function and $\ell : \mathbb{R} \to [0, \infty)$ a measurable loss. For every data point (x, y),*

$$\mathbb{E}_\theta\, \ell\big(y - f_\theta(x)\big) = \int_{\mathbb{R}} \ell(y - t)\, \nu_x^F(dt).$$

Consequently, if two random functions F and G have $\nu_x^F = \nu_x^G$ for every x, their expected losses are equal for every such ℓ and every data distribution μ.

Proof. By definition ν_x^F is the image of π under $\theta \mapsto f_\theta(x)$. The displayed identity is the change-of-variables formula for an image measure, applied to the measurable function $t \mapsto \ell(y - t)$. Integrating both sides against μ gives the second statement. □

The proof is one line, and the content is in what is absent: the joint law of $\big(f_\theta(x), f_\theta(x')\big)$ at two inputs does not enter. How f moves between inputs is what "squiggly" means, so the expected initial loss cannot register squiggliness. The lemma is about the expectation over θ only. The variance of the initial loss, the probability of a low initial loss, and everything that happens during training do depend on joint laws.

Corollary 3 (Random-phase waves)**.** *Let $g : \mathbb{R} \to \mathbb{R}$ be measurable and T-periodic, let φ be uniform on $[0, T)$, and for a frequency $w \in \mathbb{R}$ let $f_{w,\varphi}(x) = g(wx + \varphi)$ on $X = \mathbb{R}$. Then the one-point marginal of $f_{w,\cdot}$ at every x is the law of $g(\varphi)$, for every w. Hence the expected loss $\mathbb{E}_{(x,y)\sim\mu}\mathbb{E}_\varphi\, \ell\big(y - f_{w,\varphi}(x)\big)$ is the same for every w, every measurable $\ell \geq 0$ and every μ. No sequence of frequencies makes it decrease.*

Proof. The uniform distribution on the circle $\mathbb{R}/T\mathbb{Z}$ is invariant under translation, so $wx+\varphi \bmod T$ is uniform on $[0, T)$ for every fixed w and x, and $g(wx+\varphi)$ has the law of $g(\varphi)$. Apply Theorem 2. □

For the curves of Section 3, $g = p\sin$ with peak $p \leq 1$, and data with y uniform on $[-1, 1]$ independent of x. Since $\mathbb{E}_y|y - s| = (1 + s^2)/2$ for $|s| \leq 1$, the expected vertical error is $(1 + p^2/2)/2$ at every frequency.

Corollary 4 (ReLU networks under He initialisation)**.** *Let $f_\theta(x) = u^\top \sigma\big(W^L \sigma(\cdots \sigma(W^1 x))\big)$ for $x \in \mathbb{R}^d$, with σ the ReLU, no biases, hidden widths $n_1, \ldots, n_L$, $n_0 = d$, and independent weights $W_{ij}^k \sim N(0, 2/n_{k-1})$ and $u_i \sim N(0, 2/n_L)$. Then for every x and y,*

$$\mathbb{E} f_\theta(x) = 0, \qquad \mathbb{E} f_\theta(x)^2 = \frac{2\|x\|^2}{d}, \qquad \mathbb{E}\big(y - f_\theta(x)\big)^2 = y^2 + \frac{2\|x\|^2}{d},$$

whatever the depth and the widths. If all weights are instead $N(0, 1)$, the last quantity is $y^2 + \|x\|^2 \prod_{k=1}^{L} (n_k/2)$.

The proof is a layer-by-layer recursion for $\mathbb{E}\|h^k\|^2$ and is in Appendix A. It uses only that the weights are independent, symmetric and of the stated variance; preserving the second moment across layers is the design criterion of this initialisation [He et al., 2015]. In the terms of Theorem 2, the squared loss reads only the first two moments of ν_x, and these do not depend on the width. The full law ν_x does depend on the width, so for other losses the corollary gives no exact statement.

2.4 The tube identity and a bound from length

For a non-empty set S in a metric space, write $N_r(S) = \{q : \operatorname{dist}(q, S) \leq r\}$ for the closed tube of radius r around S.

Theorem 5 (Tube identity)**.** *Let S be a non-empty subset of a metric space and P a random point of that space. Then*

$$\mathbb{E}\,\operatorname{dist}(P,S) = \int_0^\infty \mathbb{P}\big(\operatorname{dist}(P,S) > r\big)\,dr = \int_0^\infty \mathbb{P}\big(P \notin N_r(S)\big)\,dr,$$

where both sides may be infinite.

Proof. The function $q \mapsto \operatorname{dist}(q,S)$ is 1-Lipschitz, hence measurable, and non-negative. For a non-negative random variable Z we have $Z = \int_0^\infty \mathbf{1}\{Z > r\}\,dr$, and Tonelli's theorem gives $\mathbb{E}Z = \int_0^\infty \mathbb{P}(Z > r)\,dr$. This is the layer-cake formula [Lieb and Loss, 2001, Section 1.13]. Take $Z = \operatorname{dist}(P,S)$. □

The identity holds for any set in any metric space. It reduces the mean distance to the volume of tubes, which for curves in the plane is controlled by length.

Lemma 6. *Let $S \subset \mathbb{R}^2$ be the image of a rectifiable curve of length l. Then the area of $N_r(S)$ is at most $2lr + \pi r^2$ for every $r \geq 0$.*

The proof, by polygonal approximation, is in Appendix A.

Proposition 7 (Length bound)**.** *Let $\Omega \subset \mathbb{R}^2$ be measurable with area $V \in (0,\infty)$, let P be uniform on Ω, and let S be the image of a rectifiable curve of length l. Then*

$$\mathbb{E}\,\operatorname{dist}(P,S) \;\geq\; \frac{V}{4l + 2\sqrt{\pi V}}.$$

Proof. Put $r_0 = V/(2l + \sqrt{\pi V})$. For $0 \leq r \leq r_0$ we have $\pi r \leq \pi V/\sqrt{\pi V} = \sqrt{\pi V}$, so by Lemma 6 the area of $N_r(S)$ is at most $(2l + \sqrt{\pi V})\,r = Vr/r_0$. Therefore $\mathbb{P}(P \notin N_r(S)) \geq 1 - r/r_0$ for $r \leq r_0$, and by Theorem 5 $\mathbb{E}\,\operatorname{dist}(P,S) \geq \int_0^{r_0}(1 - r/r_0)\,dr = r_0/2$. □

To leading order in l the bound is $V/(4l)$. We draw three consequences. Length is necessary: a mean distance of at most ε requires $l \geq V/(4\varepsilon) - \sqrt{\pi V}/2$. Length is not sufficient: a part of the curve further than r from Ω adds nothing to $N_r(S) \cap \Omega$, so length outside the data region is wasted (Section 3.2). And the bound is nearly attained only if the tube around the curve stays inside Ω and overlaps itself little, that is, if the length is spread evenly over the region. The proposition is for curves in the plane; for inputs of dimension above one the graph is a surface, and its area takes the role of length.

2.5 The sine family

The experiments use $f_{w,\varphi}(x) = p\sin(w\pi x + \varphi)$ on $x \in [-1,1]$ with φ uniform on $[0,2\pi)$, and a point P uniform on $\Omega = [-1,1]^2$, so $V = 4$. The curve makes w full periods across the box and is $p\,w\pi$-Lipschitz. Write $D_p(w)$ for the mean of δ over P and φ, and $l(w)$ for the arc length.

Proposition 8 (The sine that spans the data; leading order)**.** *For $p = 1$, as $w \to \infty$,*

$$\frac{l(w)}{w} \to 4, \qquad w\,D_1(w) \to \frac{2 - 8/\pi^2}{4} \approx 0.2974, \qquad D_1(w)\,l(w) \to 2 - \frac{8}{\pi^2} \approx 1.189.$$

The first limit is proved in Appendix A. The second is a leading-order derivation. At height y the curve crosses on two interleaved lattices of period $2/w$, with gaps $(1/w)(1 \mp 2\arcsin(y)/\pi)$. At high frequency the strokes are nearly vertical, so the nearest point of the curve is, to leading order, the nearest crossing at the same height. Averaging that distance over x and then over y, with $\int_0^1 \arcsin(y)^2\,dy = \pi^2/4 - 2$, gives the constant. The step from horizontal to Euclidean distance is argued in outline in Appendix A, without all details, and the constant is checked numerically in Section 3.3. The limit 1.189 exceeds the value $V/4 = 1$ of Proposition 7 because the strokes of a sine are not evenly spaced.

Proposition 9 (Band-edge floor)**.** *For $0 < p \leq 1$ and $w \geq 1$,*

$$\frac{(1-p)^2}{2} \;\leq\; D_p(w) \;\leq\; \frac{(1-p)^2}{2} + \frac{2}{w}.$$

The proof is in Appendix A. The curve stays in the band $|y| \le p$, so a point at height $|y| > p$ is never nearer than $|y| - p$, and $\mathbb{E}\max(0, |y| - p) = (1-p)^2/2$. Raising the frequency removes the rest at rate $O(1/w)$. The floor depends on the amplitude relative to the range of the data. It does not depend on the number of data points: $D_p(w)$ is the mean of a per-point quantity, so its expectation is the same for a sample of any size.

2.6 The covering bound and the theorem for sines

The upper bound in Proposition 9 is an instance of a general fact, which says what a function must do to be near the data.

Definition 10 (Covering)**.** A function $f : X \to \mathbb{R}$ *covers* a set of labels $S \subset \mathbb{R}$ *within* ρ if for every $x \in X$ and every $y \in S$ there is an $x' \in X$ with $d(x, x') \le \rho$ and $f(x') = y$.

The smallest such ρ is an oscillation scale: how far one must move in the input to see every label.

Theorem 11 (Covering bound)**.** *If f covers S within ρ, then $\delta_f(x, y) \le \rho$ for every data point with $y \in S$. If in addition f is K-Lipschitz, then $v_f(x, y) \le \sqrt{1 + K^2}\,\rho$ for every such point.*

Proof. Take x' as in Definition 10. The point $(x', y) = (x', f(x'))$ lies on Γ_f at distance $d(x, x') \le \rho$ from (x, y). The second statement follows from Theorem 1. □

So a small oscillation scale is sufficient for a small nearest-point distance, and by the second statement it has a price: a function that covers labels at vertical distance v within ρ has slope at least $\sqrt{(v/\rho)^2 - 1}$. Claims (G) and (P) for a family of models therefore come down to how its oscillation scale and its slope depend on its size. For sines both are explicit.

Theorem 12 (The squiggly line theorem, for sines)**.** *Let $X = \mathbb{R}$, $p > 0$ and $w > 0$, and let $f_{w,\varphi}(x) = p\sin(w\pi x + \varphi)$.*

(i) For every phase φ and every data point (x, y) with $|y| \le p$,

$$\frac{v_{f_{w,\varphi}}(x, y)}{\sqrt{1 + (p\,w\pi)^2}} \;\le\; \delta_{f_{w,\varphi}}(x, y) \;\le\; \frac{1}{w}.$$

(ii) If μ is a probability measure on data points with $|y| \le p$ almost surely and φ is uniform on $[0, 2\pi)$, the expected nearest-point distance is at most $1/w$, and tends to 0 as $w \to \infty$.

(iii) For every measurable loss $\ell \ge 0$ and every μ, the expected loss $\mathbb{E}_{(x,y)\sim\mu}\mathbb{E}_\varphi\,\ell\big(y - f_{w,\varphi}(x)\big)$ is the same for every w.

Proof. (i) The function is $p\,w\pi$-Lipschitz, which gives the lower bound by Theorem 1. For the upper bound we show that it covers $[-p, p]$ within $1/w$. Given x and y with $|y| \le p$, put $t = \arcsin(y/p)$ and $\theta_0 = w\pi x + \varphi$, let k be an integer nearest to $(\theta_0 - t)/(2\pi)$, and put $\theta = t + 2\pi k$. Then $|\theta - \theta_0| \le \pi$. The input $x' = (\theta - \varphi)/(w\pi)$ satisfies $|x - x'| = |\theta - \theta_0|/(w\pi) \le 1/w$ and $f_{w,\varphi}(x') = p\sin t = y$. Apply Theorem 11. (ii) The bound in (i) holds for every phase and almost every data point, and $\delta \ge 0$. (iii) is Corollary 3. □

Parts (i) and (ii) are claim (G), with a rate: both ends of the sandwich fall like $1/w$. Part (iii) is the failure of claim (L). On the bounded domain $[-1, 1]$ of the experiments the same argument gives $2/w$ in place of $1/w$, which is the upper bound of Proposition 9; the exact constant is the one in Proposition 8.

3 Experiments

All experiments are small numerical computations with our own scripts (Appendix D lists them). Unless stated otherwise, curves are $f(x) = A\sqrt{2}\,\sin(w\pi x + \varphi)$ on $[-1, 1]$ and data points are uniform on $[-1, 1]^2$, with y independent of x. The default amplitude is $A = 0.577$, the standard deviation of the labels, so the peak is $p = A\sqrt{2} = 0.816$. Each curve is sampled as a polyline with consecutive samples at most 0.002 apart along the curve, and the distance from a point to the polyline is computed exactly.

Table 1: Frequency isolation. Mean nearest-point distance and mean vertical error against frequency w, with the two lower bounds on the nearest-point distance: the mean vertical error divided by $\sqrt{1+K^2}$ with $K = p\,w\pi$ (Theorem 1), and $4/(4l + 4\sqrt{\pi})$ (Proposition 7 with $V = 4$).

w	arc length l	nearest-point	vertical	bound, Theorem 1	bound, Proposition 7
0.25	2.19	0.6129	0.6672	0.5617	0.2524
0.5	2.66	0.4890	0.6664	0.4099	0.2256
1	3.96	0.3095	0.6649	0.2416	0.1744
2	6.96	0.1717	0.6651	0.1273	0.1145
4	13.32	0.0928	0.6663	0.0647	0.0663
8	26.26	0.0529	0.6678	0.0325	0.0357
16	52.31	0.0338	0.6665	0.0162	0.0185
32	104.50	0.0249	0.6657	0.0081	0.0094
64	208.92	0.0207	0.6688	0.0041	0.0047

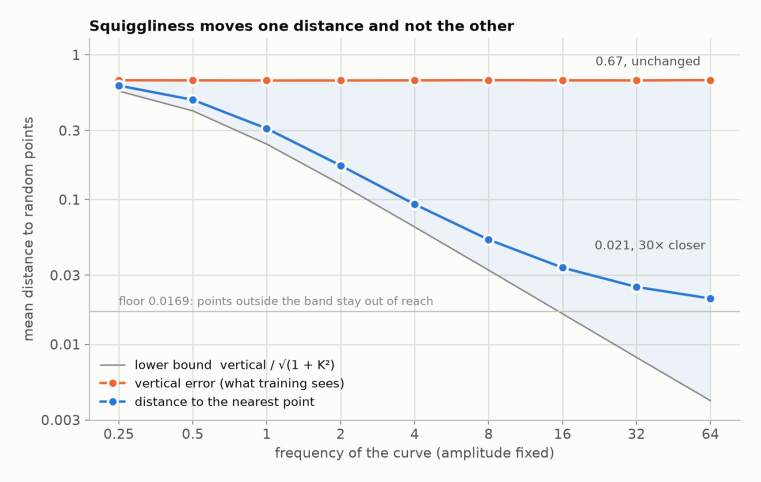


Figure 2: The data of Table 1. The nearest-point distance (blue) falls with frequency and stays above the lower bound of Theorem 1 (grey line). The vertical error (orange) does not move. The horizontal line is the floor $(1-p)^2/2 = 0.0169$ of Proposition 9.

3.1 Frequency with everything else held fixed

Earlier tests of the conjecture varied polynomial degree or network width, which changes the output amplitude together with the squiggliness. Here the frequency w is the only variable: by Corollary 3 the distribution of $f(x)$ at every x is identical in every row of Table 1. We use 3000 trials of 30 points each and 100 stratified random phases per frequency.

The nearest-point distance falls from 0.6129 to 0.0207, a factor of 29.6. Claim (G) holds, and not through amplitude. The vertical error lies between 0.6649 and 0.6688 in every row, with no trend; the value predicted after Corollary 3 is $(1+p^2/2)/2 = 0.6665$. The flat column is Corollary 3 observed, and the differences between rows are sampling noise. Both lower bounds hold in every row. At $w = 64$ the function has Lipschitz constant 164, and this steepness is what Theorem 1 requires for a nearest-point distance 32 times smaller than the vertical error. From $w = 16$ the gap to both bounds widens, because the distance approaches the floor of Proposition 9 (Figure 2).

3.2 A straight line of the same length

If length alone produced the effect, a straight segment of the same arc length should score like the sine. We take $y = mx + b$ on $x \in [-1, 1]$ with $m = \sqrt{(l/2)^2 - 1}$, so that its length is l, and b uniform on $[-1, 1]$.

Table 2: Mean nearest-point distance for the sine and for a straight segment of the same length. The last two columns give the slope of the segment and the share of its length that lies outside the data box $[-1, 1]^2$, averaged over b.

w	arc length	sine	straight segment	slope m	share outside the box
4	13.32	0.0928	0.5022	6.6	85%
16	52.31	0.0338	0.5007	26.1	96%
64	208.92	0.0207	0.5006	104.5	99%

Table 3: Knee frequency w^* against the number of points n. The prediction under test was slope 1 of $\log w^*$ on $\log n$. Confidence intervals are from a bootstrap over trials.

criterion	$n = 10$	30	100	300	1000	slope	95% interval
two-segment fit	21.6	21.6	21.6	21.6	21.6	-0.000	$[-0.004, 0.004]$
doubling gains $< 10\%$	60.1	61.7	61.4	61.0	59.4	-0.003	$[-0.014, 0.008]$

The segment stays at 0.50 while the sine reaches 0.021 (Table 2). This comparison needs a caveat. A segment of length 209 over $x \in [-1, 1]$ has slope 104, and about 99% of it lies above or below the data. The part inside the box has length about 2 in all three rows, and a near-vertical segment of length 2 through the middle of the box has mean distance $\mathbb{E}|x| = 0.5$. The table therefore shows that length outside the data region is worth nothing. It does not show that length is irrelevant: by Proposition 7, length is necessary, and the sine achieves its small distance because its length is inside the box and spread across it.

3.3 What sets the floor

An earlier version of this work predicted that the distance stops falling once the wavelength $2/w$ drops below the mean spacing $2/n$ of n data points, so that the knee of the curve $D(w)$ would sit at $w^* \propto n$. We swept $n \in \{10, 30, 100, 300, 1000\}$ and 25 frequencies in $[1, 1000]$ with $30\,000$ points per n, and located the knee by two criteria fixed in advance: the breakpoint of a continuous two-segment fit of $\log D$ on $\log w$, and the smallest w at which doubling w lowers D by less than 10%.

The slope is 0, not 1 (Table 3), and the five curves $D(w)$ agree to within 3.1% at every frequency. No experiment was needed for this: the quantity is a mean of per-point distances, so its expectation cannot depend on n. What sets the knee is the band edge of Proposition 9. With $p = 0.816$ the floor is $D_\infty = (1 - p)^2/2 = 0.0169$, and a fit of $D - D_\infty = c\, w^{-\beta}$ for $w \geq 20$ gives $\beta = 0.990$ and $c = 0.248$. For $D = D_\infty + c/w$ the 10% criterion triggers at $c(1 - 2r)/(2rD_\infty) = 58.6$ with $r = 0.1$, against 59 to 62 measured.

As a control we set $A\sqrt{2} = 1$, so that the band covers the data and the floor is zero. The saturation disappears (Figure 3): the distance follows $0.301\, w^{-1.002}$ up to $w = 1000$, where it is 0.00030, and the 10% criterion never triggers.[2] Proposition 8 predicts $w\, D_1(w) \to 0.2974$. The repository reports 0.2965 at $w = 256$; a separate run for this paper, with $40\,000$ points and 8 phases, gave 0.298 at $w = 64$ and at $w = 256$, and $D\, l = 1.192$ against the predicted 1.189.

3.4 Random networks

Initial loss against width. We draw 200 points with x uniform on $[-1, 1]^{10}$ and y uniform on $[-1, 1]$, and for each width 200 untrained ReLU networks with two hidden layers and no biases, as in Corollary 4. The prediction of Corollary 4, evaluated on the sample, is the mean of $y^2 + 2\|x\|^2/d$, which is 1.03.

A wider network does not start closer (Table 4; Figure 4 in Appendix D). The measured values scatter around the prediction; the largest deviation, at width 256, is 1.9 standard errors. With unscaled weights a wider network starts much further away, because its output grows with width. That growth is an amplitude effect, and it is the confound that the design of Section 3.1 removes.

[2] The two-segment fit still reports a breakpoint, at 1.8, because it always returns one.

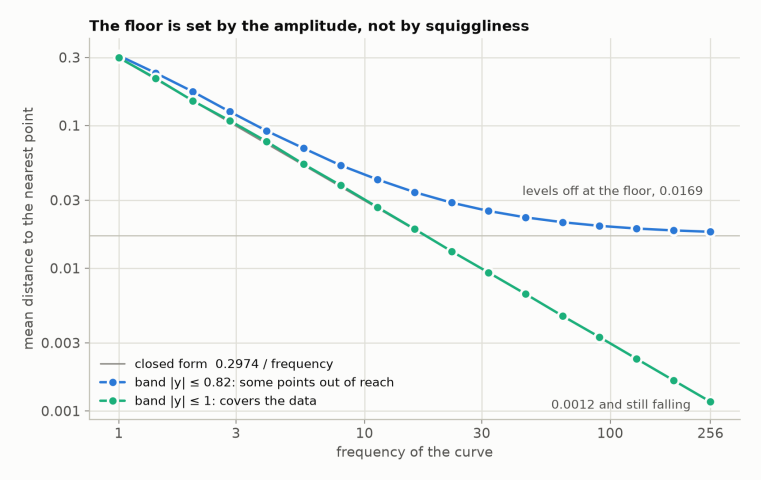


Figure 3: The floor comes from the amplitude. Blue: default amplitude, band $|y| \leq 0.82$; the distance levels off at 0.0169. Green: band $|y| \leq 1$; no floor, and the distance follows the constant of Proposition 8 (grey line, under the green one).

Table 4: Mean squared error before training, 200 networks per width ($\pm$ one standard error). Predictions are from Corollary 4.

	He initialisation		$N(0,1)$ weights	
width	measured	predicted	measured	predicted
4	1.07 ± 0.12	1.03	15	14
16	1.02 ± 0.05	1.03	220	219
64	1.07 ± 0.04	1.03	3.45×10^3	3.51×10^3
256	1.10 ± 0.04	1.03	6.02×10^4	5.61×10^4
1024	1.03 ± 0.04	1.03	8.23×10^5	8.97×10^5

Curve length against width and depth. For claim (P) we look at the curves themselves, for a one-dimensional input. Each network has He-uniform weights and hidden biases uniform on $[-0.5, 0.5]$, and its output is shifted and scaled so that its range over $[-1, 1]$ is exactly $[-1, 1]$. We measure arc length, the number of turns (sign changes of the slope), and the mean nearest-point distance to 200 fixed random points, over 300 networks per size.

Width adds turns and no length (Table 6 in Appendix D). From width 2 to 512, with two hidden layers, the mean number of turns rises from 0.5 to 15, the arc length stays between 3.4 and 3.6, and the nearest-point distance is flat from width 16, between 0.42 and 0.43. Depth adds some length: from 1 to 16 layers at width 64 the arc length rises from 3.4 to 5.1 and the distance falls from 0.44 to 0.34. For comparison, the same code gives length 4.61 and distance 0.351 for a sine with one period across the box, and 64.07 and 0.021 for a sine with 16 periods. Even at depth 16 the random curves resemble the first and not the second. So claim (P) fails for width and holds weakly for depth.

This measurement has caveats. The input is one-dimensional. Rescaling the output controls the amplitude but does not hold the marginals of $f(x)$ fixed, unlike Section 3.1. Only the nearest-point distance is measured, not the vertical error. The distances use a fixed grid of 4001 samples, which overestimates the distance by about 7% for the sine with 16 periods (0.0209 against 0.0196 with exact projection). Finally, the rescaling and the biases mean that the depth trend is not directly comparable with the theorem of Hanin et al. [2022] discussed in Section 6.

4 Where "starts closer" is true: parameter space

Theorem 2 shows that a larger random model does not start with a smaller loss. It can still start nearer to a fit, if nearness is measured in the space of parameters. For models that are linear in their

parameters this is elementary, and it is the lazy-training picture of Jacot et al. [2018], Chizat et al. [2019] in its simplest case. We state it because it is the form in which the original idea holds.

4.1 Setting

Let Θ and Y be real Hilbert spaces and $A : \Theta \to Y$ a bounded linear map. We read $\theta \in \Theta$ as the parameters, $A\theta$ as the outputs of the model on the data, and $y \in Y$ as the labels. The *residual* is $r(\theta) = y - A\theta$, a *fit* is a θ with $A\theta = y$, and the *travel* from θ_0 is the distance to the nearest fit,

$$\tau(\theta_0) = \inf\{\|\theta - \theta_0\| : A\theta = y\}.$$

A Fourier series is such a model in its coefficients, and so is a neural network in the weights of its last layer when the earlier layers are held fixed. The *kernel* is $K = AA^*$.

4.2 Results

Proposition 13 (One loss curve)**.** *If $A\theta_1 = y$, then $r\big(\theta_0 + t(\theta_1 - \theta_0)\big) = (1 - t)\, r(\theta_0)$ for every $t \in \mathbb{R}$. Consequently, if $d = \|\theta_1 - \theta_0\| > 0$ and $0 \le \varepsilon \le d$, there is a θ with $\|\theta - \theta_0\| = \varepsilon$ and $\|r(\theta)\| = (1 - \varepsilon/d)\, \|r(\theta_0)\|$.*

Proof. $y - A\theta_0 - t(A\theta_1 - A\theta_0) = (1 - t)(y - A\theta_0)$. For the second statement take $t = \varepsilon/d$. □

Every model walks the same loss curve from its start to a fit; what differs is the length of the walk. A fit at distance d means that a movement of ε removes the fraction ε/d of the residual.

Theorem 14 (A model is its kernel)**.** (a) *One step of gradient descent on $\frac{1}{2}\|y - A\theta\|^2$ with step size η, $\theta^+ = \theta + \eta A^* r(\theta)$, gives $r(\theta^+) = (I - \eta K)\, r(\theta)$.* (b) *If K is invertible, then $\tau(\theta_0) = \|A^* K^{-1} r(\theta_0)\| = \sqrt{\langle K^{-1} r(\theta_0), r(\theta_0)\rangle}$. Hence two models with the same kernel and the same initial residual have the same residual after every gradient step, and the same travel.*

Proof. (a) $y - A\theta - \eta AA^* r(\theta) = r(\theta) - \eta K r(\theta)$. (b) Write $r_0 = r(\theta_0)$ and $\delta^* = A^* K^{-1} r_0$. Then $A\delta^* = r_0$, so $\theta_0 + \delta^*$ is a fit. If $\theta_0 + \delta$ is any fit, then $A\delta = r_0$ and

$$\|\delta^*\|^2 = \langle K^{-1} r_0, AA^* K^{-1} r_0\rangle = \langle K^{-1} r_0, r_0\rangle = \langle K^{-1} r_0, A\delta\rangle = \langle \delta^*, \delta\rangle \le \|\delta^*\|\, \|\delta\|,$$

so $\|\delta^*\| \le \|\delta\|$. The second and third expressions show that $\|\delta^*\|^2 = \langle K^{-1} r_0, r_0\rangle$, which involves only K and r_0. □

Corollary 15 (The exchange rate)**.** *If $K = c\, I$ with $c > 0$, that is, if the features are orthogonal on the data side with squared length c, then*

$$\tau(\theta_0) = \frac{\|r(\theta_0)\|}{\sqrt{c}},$$

the residual after t gradient steps is $(1 - \eta c)^t\, r(\theta_0)$, and with $\eta = 1/c$ a single step reaches a fit. For $c = 1$ the travel equals the size of the residual.

Loss and travel are then one quantity in two units, and the number of parameters enters only through the rate $\sqrt{c}$ between them.

Proposition 16 (More parameters never move the fit away)**.** *Let $B : \Theta' \to Y$ be a second bounded linear map and consider the larger model $(\theta, \theta') \mapsto A\theta + B\theta'$. If the smaller model has a fit, the travel of the larger model from $(\theta_0, 0)$ is at most the travel of the smaller model from θ_0.*

Proof. If $A\theta = y$ then $(\theta, 0)$ is a fit of the larger model at the same distance from $(\theta_0, 0)$. □

Together with Theorem 2 this gives the corrected form of the conjecture, for models linear in their parameters: with no movement allowed, a larger random model is no closer to the data; with any positive amount of movement allowed, it is at least as close. Proposition 16 starts the new parameters at zero. The experiment below starts them at random.

Table 5: A random Fourier series with P parameters on 20 points with random labels. Initial squared error and nearest-point distance are means; travel and steps are medians. The last column is $\sqrt{n \cdot \mathrm{MSE}/(m+1)}$, the travel that Corollary 15 gives if the features were exactly orthogonal with $c = m + 1$. The nearest-point distance was not measured above $P = 129$.

P	initial MSE	nearest-point	travel	gradient steps	$\sqrt{n\,\mathrm{MSE}/(m+1)}$
9	0.65	0.229	no fit	never	–
17	0.71	0.157	no fit	never	–
21	0.65	0.128	448	1.9×10^8	–
25	0.64	0.110	35.5	7.9×10^4	–
33	0.66	0.090	16.0	3.3×10^4	0.88
65	0.71	0.056	1.61	166	0.66
129	0.66	0.027	0.48	13	0.45
257	0.66	–	0.34	3	0.32
513	0.63	–	0.23	3	0.22

4.3 A family in which the number of parameters is the only dial

The sine family has one random number, so it cannot test a statement about the number of parameters. We therefore take a Fourier series with m terms,

$$f_\theta(x) = a_0 + \sum_{k=1}^{m} \big(a_k \cos(k\pi x) + b_k \sin(k\pi x)\big),$$

with $P = 2m + 1$ parameters drawn independently from $N(0, V/(m+1))$, $V = 1/3$. The output then has variance V at every x and every m, so the one-point marginals do not depend on m. The data are $n = 20$ points with x and y uniform on $[-1, 1]$; a fit exists once $P \geq n$. For each m we use 20 data sets and 20 random starts, and measure the initial squared error, the nearest-point distance, the travel τ, and the number of gradient steps needed to reach a mean squared error of 10^{-3} with the step size $1/\lambda_{\max}(K)$.

Table 5 shows three things. The initial squared error does not move: it scatters around $\mathbb{E}y^2 + V = 2/3$, as Theorem 2 requires. The travel and the number of steps fall steeply: from 21 to 513 parameters the travel falls by a factor of about 2000, and training goes from 1.9×10^8 steps to 3. And the worst case is at the threshold $P \approx n$, where a fit exists but is far away; this is where double descent has its peak [Belkin et al., 2019, Nakkiran et al., 2020].

Far above the threshold the measured travel agrees with Corollary 15. The kernel of these features has diagonal entries $m+1$ and, for scattered points and $m \gg n$, comparatively small off-diagonal entries, so it is close to $(m+1)I$. The last column of Table 5 is within 7% of the measured travel from $P = 129$ upwards, and travel falls like $1/\sqrt{P}$. Near the threshold the features are far from orthogonal and the formula fails by a wide margin. That the kernel is close to $(m+1)I$ is an observation here, not a proved statement.

The nearest-point distance also falls, roughly like $1/P$. This is Theorem 1 again: a random series with more terms of this scaling is steeper. So in this family claims (G) and (P) both hold, and (L) still fails. A larger random model is not nearly right. It is nearly adjustable.

5 Formalisation

The repository contains a Lean 4 project [de Moura and Ullrich, 2021] on Mathlib [The mathlib Community, 2020]. A randomly initialised model is a parameter type with a probability measure and an evaluation map, and one definition says that the expected cost is strictly decreasing in the size of the model; the conjecture and the learning claim are that definition for the nearest-point distance and for a loss of the residual.

Proved: Theorem 1, for inputs in any seminormed group; Theorem 5, for an integrable distance in any pseudo-metric space; Theorem 2; Corollary 3, from the translation invariance of Haar measure on the circle; Theorem 11, pointwise and in expectation; Theorem 12, stated for $A\sin(wx + \varphi)$ with

the bound π/w, which is the statement above after replacing w by $w\pi$; and Propositions 13 and 16, Theorem 14, and Corollary 15, for bounded linear maps between complete real inner-product spaces, with "K invertible" taken as the existence of a right inverse. The statements of Appendix B about ReLU chains and polarizers are proved as well.

Not formalised: Lemma 6 and Propositions 7 to 9; Corollary 4, so the width of a network enters the formal development nowhere; and the observation of Section 4.3 that the kernel of random Fourier features is close to a multiple of the identity.

The project contains no `sorry`, every theorem depends only on Lean's three standard axioms, and a workflow in the repository rebuilds the project on every change. A proof assistant checks that the proofs follow from the definitions. It does not check that the definitions say what the prose says, and we have tried to keep them short enough to be read. Appendix E lists the Lean names.

6 Related work

We describe each work only as far as its abstract does; the reading of how it bears on the conjecture is ours.

Length of random network curves (claim P). Raghu et al. [2017] propose trajectory length, the length of the image of a one-dimensional input path, as a measure of expressivity, and report that it grows exponentially with depth. Hanin et al. [2022] prove that for ReLU networks with standard random initialisation the expected length distortion does not grow with depth. Hanin and Rolnick [2019] bound the average number of activation patterns of a ReLU network at initialisation in terms of the number of neurons and the input dimension. By Proposition 7, a curve that does not get longer cannot get much closer in the nearest-point sense.

Squiggliness after training. Dherin et al. [2021] argue that stochastic gradient descent implicitly favours functions of low geometric complexity, which for one-dimensional regression is the arc length. Bubeck and Sellke [2021] show that, for broad classes of models and data, fitting the data smoothly requires d times more parameters than merely interpolating it, where d is the input dimension; in that account parameters buy smoothness, the opposite of (P). Humayun et al. [2024] report that the linear regions of a network migrate away from the training samples late in training, and Hosseini and Fedorenko [2023] that trained language models straighten the trajectory of a sentence across layers while untrained ones do not.

Size helping before or without training. Frankle and Carbin [2019] find sparse subnetworks that train in isolation to the accuracy of the full network. Ramanujan et al. [2020] show that a random network with fixed weights contains subnetworks that perform well untrained, and that these improve as the network gets wider and deeper. Larsen et al. [2022] find that training within a random subspace succeeds once its dimension passes a threshold that depends on the target loss and the initial loss. Chiang et al. [2023] report that sampling random parameter vectors until one fits the training set reaches test accuracy comparable to gradient descent in low-sample regimes. These are statements about subnetworks or about weight space, not about the nearest-point distance of the full random function, and Corollary 4 is compatible with them. Martinelli et al. [2026] argue that the lottery-ticket analogy does not explain overparameterization and propose that width adds dimensions along which optimisation escapes bad minima.

Wide networks. In the lazy regime a model behaves like its linearisation around the initial weights [Chizat et al., 2019], with the neural tangent kernel governing training [Jacot et al., 2018], and wide random networks converge to Gaussian processes [Matthews et al., 2018]. This is the setting in which "starts closer" is true, and Section 4 is its simplest case: a model that is exactly linear in its parameters, where the kernel does not move during training.

7 Limitations

One-dimensional inputs. The geometry experiments and Propositions 7 to 9 are for curves in the plane; claim (G) has not been measured for surfaces. *A degenerate family.* For fixed w the functions

$p\sin(w\pi x+\varphi)$ span a two-dimensional space, and their covariance kernel $\frac{p^2}{2}\cos(w\pi(x-x'))$ has rank 2. No member can fit more than two generic points at once, at any frequency, and the mean per-point distance cannot see this because it scores each point alone. *Nothing to learn.* The labels are independent of the inputs, so the experiments concern the loss at initialisation only and say nothing about learning speed or generalisation. *Expectation only.* Theorem 2 and Corollary 4 are about the mean initial loss; whether the probability of a low initial loss changes with width is not addressed. *Own scripts.* The experiments are small computations written for this question, not an established benchmark, and the code has needed correction more than once (Appendix C). *Linear models only.* Section 4 is for models that are linear in their parameters. A network trained in every layer is not, and its kernel moves during training; nothing here bounds by how much. *Networks.* For random networks claim (G) is measured in one dimension only, and the oscillation scale of Theorem 11 is not computed for them. *Proof status.* The constant in Proposition 8 rests on a leading-order argument and a numerical check, and the near-orthogonality used in Section 4.3 is an observation.

8 Discussion

The conjecture joined two facts that are each true: a squigglier curve is nearer to random points, and larger networks learn faster. The link between them fails at both ends. Width does not make a random network's curve longer, and the nearness that squiggliness buys is nearness to points displaced in the input direction, which a regression loss does not reward and which Theorem 1 shows is paid for in slope.

What replaces the link. A larger model starts nearer to a fit in parameter space, not in loss (Section 4). The two are related by Proposition 13: the loss to be removed is the same at every size, and the distance over which it is removed shrinks. For models linear in their parameters this is exact and is governed by the kernel.

What remains a conjecture. For neural networks, claim (G) asks how the oscillation scale of a random network depends on its width, its depth and the scale of its weights. Theorem 11 and Theorem 1 bound the nearest-point distance between the slope and the oscillation scale, and the results cited in Section 6 suggest that neither improves with width at standard initialisation. We do not prove this.

An untested proposal: joint fit. The nearest-point distance scores each point alone. A cost that sees all points at once is $y^\top K^{-1}y$, where $y\in\mathbb{R}^n$ collects the labels and K is the positive-definite covariance matrix of the random function at the n inputs; it is the data-fit term of the Gaussian-process marginal likelihood [Rasmussen and Williams, 2006]. For uncorrelated labels of zero mean and unit variance its mean is $\operatorname{tr}(K^{-1})$, and $\operatorname{tr}(K^{-1})\ge n^2/\operatorname{tr}(K)$ with equality if and only if K is a multiple of the identity, by the Cauchy–Schwarz inequality $n^2=\big(\sum_i \lambda_i^{1/2}\lambda_i^{-1/2}\big)^2\le\sum_i\lambda_i\sum_i\lambda_i^{-1}$ for the eigenvalues λ_i of K. At fixed output variance $\operatorname{tr}(K)$ is fixed, so for random labels the cost is smallest for the prior whose values at different inputs are uncorrelated, the squiggliest one; for labels that vary smoothly with the input we expect the order to reverse. This is a direction only. Beyond the inequality it is untested here: we have not measured this cost for networks and do not know whether it predicts anything about training.

Conclusion. The squiggly line picture is correct as geometry, and for sines it is a theorem. It does not explain why overparameterized networks learn, because the distance it uses is not the one that training measures. The statement that does hold, that a larger model starts nearer to a fit, is a statement about parameters.

Acknowledgements

This paper was drafted with the assistance of Claude, an AI system made by Anthropic, which also wrote the Lean proofs and the experiment scripts under the author's direction. The author is responsible for the content.

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A Proofs

A.1 Second moments under He initialisation

We prove Corollary 4. Put $h^0 = x$ and $h^k = \sigma(W^k h^{k-1})$. Given h^{k-1}, the coordinates of $W^k h^{k-1}$ are $N(0, s_k^2 \|h^{k-1}\|^2)$, where s_k^2 is the weight variance of layer k. For a centred Gaussian z, symmetry gives $\mathbb{E}\,\sigma(z)^2 = \frac{1}{2}\mathbb{E}z^2$. Hence $\mathbb{E}\big[\|h^k\|^2 \mid h^{k-1}\big] = \frac{1}{2} n_k s_k^2 \|h^{k-1}\|^2$. With $s_k^2 = 2/n_{k-1}$ this reads $\mathbb{E}\|h^k\|^2/n_k = \mathbb{E}\|h^{k-1}\|^2/n_{k-1}$, so $\mathbb{E}\|h^L\|^2/n_L = \|x\|^2/d$. Given h^L, the output is $N(0, (2/n_L)\|h^L\|^2)$, so $\mathbb{E}f_\theta(x) = 0$ and $\mathbb{E}f_\theta(x)^2 = 2\,\mathbb{E}\|h^L\|^2/n_L = 2\|x\|^2/d$. The third identity follows by expanding the square. With $s_k^2 = 1$ the recursion is $\mathbb{E}\|h^k\|^2 = \frac{1}{2} n_k \mathbb{E}\|h^{k-1}\|^2$ and $\mathbb{E}f_\theta(x)^2 = \mathbb{E}\|h^L\|^2$. □

A.2 Area of a tube around a curve

We prove Lemma 6.

Polygonal paths. Let Q be a polygonal path with vertices $q_0, \dots, q_k$ and segments of lengths $s_1, \dots, s_k$. For a segment $[a, b]$, the tube $N_r([a, b])$ is a rectangle of sides $|b - a|$ and $2r$ with a half-disc of radius r at each end. Its area is $2r|b - a| + \pi r^2$, and it contains the closed disc $\bar{B}_r(a)$, so $N_r([a, b]) \setminus \bar{B}_r(a)$ has area $2r|b - a|$. Let Q_j be the path up to q_j. Then $N_r(Q_0) = \bar{B}_r(q_0)$ has area πr^2, and $N_r(Q_j) = N_r(Q_{j-1}) \cup N_r([q_{j-1}, q_j])$. Since $\bar{B}_r(q_{j-1}) \subset N_r(Q_{j-1})$, passing from Q_{j-1} to Q_j adds at most the area of $N_r([q_{j-1}, q_j]) \setminus \bar{B}_r(q_{j-1})$, which is $2r s_j$. By induction the area of $N_r(Q)$ is at most $\pi r^2 + 2r \sum_j s_j$.

Rectifiable curves. Let $\gamma : [0, 1] \to \mathbb{R}^2$ be continuous with length $l < \infty$ and image S. Fix $\varepsilon > 0$. By uniform continuity there is a partition $0 = t_0 < \cdots < t_k = 1$ such that each arc $\gamma([t_{j-1}, t_j])$ has diameter at most ε. Let Q be the polygonal path through the points $\gamma(t_j)$; its length is at most l by the definition of length. Every point of S is within ε of a vertex of Q, and S is compact, so $N_r(S) \subset N_{r+\varepsilon}(Q)$. By the first step the area of $N_r(S)$ is at most $2l(r + \varepsilon) + \pi(r + \varepsilon)^2$. Let $\varepsilon \to 0$. □

A.3 Arc length of the sine

For $f(x) = \sin(w\pi x + \varphi)$ on $[-1, 1]$ the arc length is $l(w) = \int_{-1}^{1} \sqrt{1 + w^2\pi^2 \cos^2(w\pi x + \varphi)}\, dx$. Since $c \le \sqrt{1 + c^2} \le 1 + c$ for $c \ge 0$,

$$I \le l(w) \le 2 + I, \qquad I = w\pi \int_{-1}^{1} |\cos(w\pi x + \varphi)|\, dx = \int_{\varphi - w\pi}^{\varphi + w\pi} |\cos u|\, du.$$

The function $|\cos|$ has period π and integral 2 over a period. An interval of length $2\pi w$ contains $\lfloor 2w \rfloor$ full periods and a remainder shorter than one period, so $4w - 2 \le I \le 4w + 2$. Hence $4w - 2 \le l(w) \le 4w + 4$ and $l(w)/w \to 4$. With peak p the same argument gives $l(w)/w \to 4p$, in agreement with Table 1 ($4 \times 0.816 \times 64 = 208.9$).

A.4 The constant for the sine

We derive the second limit of Proposition 8.

Take $p = 1$ and fix a phase φ and a height $y_0 \in (-1, 1)$. Put $a = \arcsin(y_0) \in (-\pi/2, \pi/2)$.

Crossings. The curve is at height y_0 where $w\pi x + \varphi$ equals a or $\pi - a$ modulo 2π. These are two lattices in x, each of period $2/w$, and consecutive crossings are separated alternately by

$$g_1 = \frac{1}{w}\Big(1 - \frac{2a}{\pi}\Big), \qquad g_2 = \frac{1}{w}\Big(1 + \frac{2a}{\pi}\Big), \qquad g_1 + g_2 = \frac{2}{w}.$$

Horizontal distance. Let $h(x_0, y_0)$ be the distance from x_0 to the nearest crossing at height y_0. A point x_0 falls in a gap of length g with probability proportional to g, and its mean distance to the nearer end is then $g/4$. Averaging over x_0 across whole periods,

$$\frac{g_1^2 + g_2^2}{4(g_1 + g_2)} = \frac{1}{4w}\Big(1 + \frac{4a^2}{\pi^2}\Big).$$

The interval $[-1,1]$ contains whole periods and end pieces of total length at most $4/w$, on which $h \le 2/w$ for $w \ge 1$. So the mean of h over x_0 uniform on $[-1,1]$ differs from the display by $O(1/w^2)$.

Average over heights. For y_0 uniform on $[-1,1]$, with $y = \sin t$,

$$\mathbb{E}\,\arcsin(y_0)^2 = \int_0^1 \arcsin(y)^2\,dy = \int_0^{\pi/2} t^2 \cos t\,dt = \Big[t^2\sin t + 2t\cos t - 2\sin t\Big]_0^{\pi/2} = \frac{\pi^2}{4} - 2.$$

Therefore $w\,\mathbb{E}h \to \frac{1}{4}\big(1 + \frac{4}{\pi^2}(\frac{\pi^2}{4} - 2)\big) = \frac{1}{4}\big(2 - \frac{8}{\pi^2}\big)$.

From horizontal to Euclidean distance (outline). It remains to show that $w\,\mathbb{E}\,\delta$ and $w\,\mathbb{E}h$ have the same limit. Clearly $\delta \le h$. For the other direction fix $\eta \in (0,1)$ and suppose $|y_0| \le 1 - \eta$. Each monotone stroke of the curve is the graph of x as a function of the height y, with derivative $1/(w\pi\sqrt{1-y^2})$. On the window of heights within $2/w$ of y_0 this derivative is at most $\kappa = 1/\big(w\pi\sqrt{1-(1-\eta+2/w)^2}\big)$ once $2/w < \eta$, and $\kappa \to 0$ as $w \to \infty$. Points of the curve outside the window are further than $2/w \ge h$ from (x_0, y_0). Inside the window, Theorem 1 applies to each stroke with the roles of x and y exchanged, and gives $\delta \ge h/\sqrt{1+\kappa^2}$. The points with $|y_0| > 1 - \eta$ have probability η and $\delta \le h \le 2/w$, so they change $w\,\mathbb{E}\,\delta$ and $w\,\mathbb{E}h$ by at most 2η each. Letting $w \to \infty$ and then $\eta \to 0$ gives the claim. We have not written out the measurability and edge details of this last step, and for that reason label the result as leading-order; the numerical check is in Section 3.3.

A.5 The band-edge floor

We prove Proposition 9.

Lower bound. The curve lies in the band $|y| \le p$. A point (x_0, y_0) is therefore at distance at least $\max(0, |y_0| - p)$ from it. For y_0 uniform on $[-1,1]$, $\mathbb{E}\max(0,|y_0|-p) = \int_p^1 (t-p)\,dt = (1-p)^2/2$.

Upper bound. We use the following fact: if $\Lambda \subset \mathbb{R}$ is a lattice of period $2/w$ with $w \ge 1$, every $x_0 \in [-1,1]$ has a point of $\Lambda \cap [-1,1]$ within $2/w$. Indeed, let $x_L \le x_0 < x_R = x_L + 2/w$ be the neighbours of x_0 in Λ. If $x_L \ge -1$ take x_L. Otherwise $x_R < -1 + 2/w \le 1$ and $x_R > x_0 \ge -1$, so take x_R.

If $|y_0| \le p$, the crossings of the curve at height y_0 contain a lattice of period $2/w$, so there is a point of the curve at height y_0 within $2/w$ of x_0, and the distance is at most $2/w$. If $y_0 > p$, the maxima of the curve form a lattice of period $2/w$ at height p, so the distance is at most $\sqrt{(y_0-p)^2 + (2/w)^2} \le (y_0 - p) + 2/w$. The case $y_0 < -p$ is the same with the minima. In every case the distance is at most $\max(0, |y_0| - p) + 2/w$, for every phase. Taking expectations gives the upper bound. □

B ReLU chains and polarizers

The statements in this appendix are proved in Lean. The main results do not use them. They concern claim (P) for the simplest ReLU networks, and an analogy with polarizers that explains why a turn needs more than one neuron.

One neuron per layer. The network $\sigma\big(w_1\sigma(w_2 \cdots \sigma(w_L x + b_L)\cdots) + b_1\big)$ is monotone, for every depth and all weights and biases: each layer is a non-decreasing or non-increasing function of its input, and a composition of such functions is again one. Such a network never turns.

Two neurons per layer. Let $T(x) = 2\sigma(x) - 4\sigma(x - \frac{1}{2})$, one layer with two neurons, which folds $[0,1]$ in half. For every L and every integer $0 \le k \le 2^L$, the L-fold composition satisfies $T^L(k/2^L) = 0$ if k is even and 1 if k is odd. So L layers of two neurons draw a curve that turns $2^L - 1$ times. Moreover, for $x \in [0,1]$,

$$\sin^2\big(\tfrac{\pi}{2}\,T^L(x)\big) = \sin^2\big(2^L \cdot \tfrac{\pi}{2}\,x\big),$$

so that, seen through the fixed function $\sin^2(\frac{\pi}{2}\,\cdot)$, L folds are a single Fourier mode of frequency 2^L. Depth in such a network plays the role of the logarithm of frequency in Section 4.3.

ReLU as a projection. In the plane, the coordinatewise ReLU sends a point to the nearest point of the positive quadrant. Let R_θ be the rotation by θ and $r \geq 0$. A layer that rotates and then applies the ReLU acts on a point on the edge of the quadrant by

$$\sigma\big(R_{-\theta}(r,0)\big) = \begin{cases} (r\cos\theta,\, 0) & 0 \leq \theta \leq \pi/2, \\ (0,0) & \pi/2 \leq \theta \leq \pi, \end{cases}$$

so N such layers with $\theta \leq \pi/2$ keep the length $r\cos^N\theta$.

Polarizers. A linear polarizer with axis at angle ϕ is the orthogonal projection P_ϕ onto that direction. Writing u_ψ for the unit vector at angle ψ, we have $P_\phi(a\,u_\psi) = a\cos(\phi-\psi)\,u_\phi$ (Malus's law for the amplitude). After k plates at angles $\theta, 2\theta, \ldots, k\theta$ the vector $a\,u_0$ has become $a\cos^k\theta\;u_{k\theta}$. A single plate at 90° delivers nothing, a plate at 45° followed by one at 90° delivers half the length, and a quarter turn made in $N \geq 2$ equal steps delivers $\cos^N(\pi/2N) \geq 1 - \pi^2/(8N)$. A turn made in many small steps loses little, and a turn made in one step loses everything. The rotate-then-ReLU layer above obeys the same rule, and a network with one neuron per layer has no intermediate direction to turn through. The comparison stops there: a stack of projections onto lines is a linear map, and a stack of ReLU layers is not.

Chains without memory. For any stack of maps applied in turn, two inputs that reach the same state stay together, the number of distinct states of a finite set of inputs never increases, and a state that every layer fixes is never left.

C What went wrong in the first version

The first scripts, written in August 2025, appeared to support the conjecture, and a draft paper reported that support. Three of their measurements were not what they were described as. The repository keeps the scripts unchanged, with a warning, and a script that reproduces each defect.

1. *The "distance" was a neuron count.* The core metric compared the label with the number of active neurons of the network, which is about half the width. It measured 4.9 at width 10 and 80.1 at width 160. It grows linearly with width, the opposite of the conjecture, and was captioned as support for it. Its apparent independence of the input dimension was the absence of any dependence on the data.
2. *The scaling law was the output fading.* A power law with exponent about 0.42 was reported for $\|\mathrm{net}(x) - x\|$, the distance from the output of a network to its own input, on a $2 \to w \to w \to 2$ ReLU stack with Xavier initialisation [Glorot and Bengio, 2010]. In that setup the output shrinks towards zero as the width grows, so the metric converges to $\|x\|$, the score of a network that returns nothing: 1.3186 at width 2 and 1.2325 at width 128, against 1.2299 for the zero function.
3. *The one-dimensional case measured against a diagonal.* One coordinate of each random point was used as both x and y, which placed every point on the line $y = x$.

The original derivation also estimated the mean distance as a power of the uncovered fraction of the region and drew a conclusion about diminishing returns that the formula does not support; Theorem 5 replaces that estimate.

Two later corrections were numerical and concern the experiments reported here.

1. *Fixed sampling grid (corrected 19 September 2026).* Curves were first sampled on a fixed grid of 4000 points. At $w = 64$ the steepest slope of the sine is about 164, so consecutive samples were about 0.08 apart vertically, more than the distances being measured. The nearest-point distance was overestimated from $w \approx 32$ upwards (0.0279 reported at $w = 64$, against 0.0207), which produced a spurious plateau. The grid is now sized for each frequency.
2. *Wrong stroke (corrected 11 November 2026).* The replacement refined the distance only on the two segments next to the nearest sample. Once the gap $1/w$ between neighbouring strokes is below the sample spacing 0.002, the nearest sample can lie on the wrong stroke.

In the control of Section 3.3 this overestimated the mean distance by 11% at $w = 1000$ (0.000324 against 0.000292). The routine now projects onto every segment that can contain the nearest point, is exact for the polyline, and agrees with a brute-force reference to better than 10^{-6} from $w = 0.25$ to $w = 1000$ at both amplitudes. Tables 1 and 2 did not change in the digits shown. The tail fits of Section 3.3 moved slightly: before the correction they read $\beta = 0.978$, $c = 0.236$, a predicted knee of 55.7, and $0.285\, w^{-0.989}$ for the control.

D Experimental details

All scripts are in the `scripts/` folder of the repository and use fixed random seeds.

- `frequency_isolation.py`: Tables 1 and 2. 3000 trials of 30 points; 100 stratified phases per frequency; about ten seconds.
- `saturation_scaling.py`: Table 3 and the fits of Section 3.3. $30\,000$ points per n, 16 phases per frequency, 25 frequencies logarithmically spaced in $[1, 1000]$, 300 bootstrap resamples; the control uses $n = 100$. Several minutes.
- `initial_mse.py`: Table 4.
- `random_relu_length.py`: Table 6.
- `fourier_params.py`: Table 5. 20 data sets and 20 random starts for each number of terms; the number of gradient steps is computed exactly in the eigenbasis of the kernel.
- `curves.py`: the sine geometry and the exact point-to-polyline distance shared by the first two scripts.
- `audit_legacy_metrics.py`: reproduces the three defects of Appendix C.
- `readme_figures.py`: draws Figures 1 to 4 from the same functions.

The numbers in Tables 1 to 4 and 6 were reproduced by running these scripts for this paper. The slope and share-outside columns of Table 2, the bound columns of Table 1 and the check of Proposition 8 quoted in Section 3.3 were computed for this paper from the same functions. Over the 1600 networks that the last script pools across sizes, the correlation between arc length and nearest-point distance is -0.76.

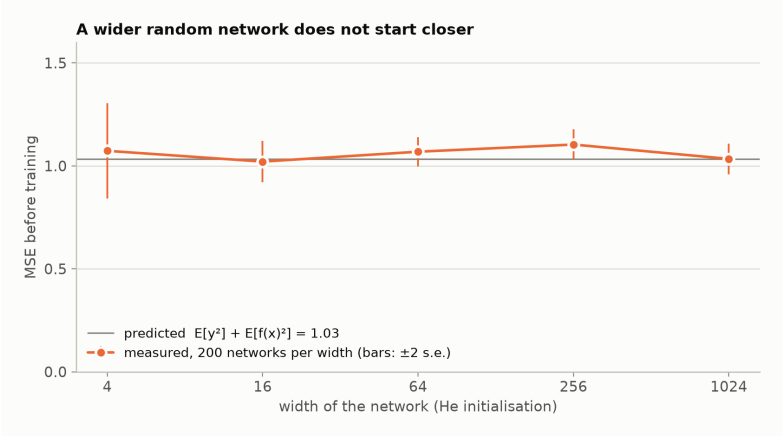


Figure 4: Initial squared error of random ReLU networks under He initialisation (Table 4), with the prediction of Corollary 4. Bars are ± 2 standard errors.

E Lean names

The Lean project is in the folder `lean/` of the repository, built with Lean 4.36.0-rc1 and the Mathlib commit recorded in `lean/lake-manifest.json`. Each file is `lean/SquigglyLine/<name>.lean`.

Table 6: Random ReLU networks with a one-dimensional input: all sizes ($\pm$ one standard error over 300 networks; no training). Setup and caveats as in Section 3.4. Reference curves measured by the same code: the diagonal $y = x$ has length 2.83 and distance 0.520; a sine with one period across the box, 4.61 and 0.351; a sine with 16 periods, 64.07 and 0.021.

width	hidden layers	parameters	arc length	turns	nearest-point
2	2	12	3.446 ± 0.040	0.47	0.4808 ± 0.0052
4	2	32	3.571 ± 0.037	1.47	0.4540 ± 0.0050
8	2	96	3.512 ± 0.034	2.42	0.4398 ± 0.0049
16	2	320	3.604 ± 0.035	3.16	0.4231 ± 0.0046
32	2	1152	3.571 ± 0.034	4.39	0.4269 ± 0.0046
64	2	4352	3.559 ± 0.035	5.89	0.4269 ± 0.0045
128	2	16896	3.636 ± 0.037	8.18	0.4179 ± 0.0046
256	2	66560	3.577 ± 0.036	11.54	0.4286 ± 0.0047
512	2	264192	3.599 ± 0.034	15.11	0.4257 ± 0.0046
64	1	192	3.417 ± 0.034	3.05	0.4428 ± 0.0047
64	2	4352	3.613 ± 0.037	5.90	0.4256 ± 0.0047
64	3	8512	3.761 ± 0.037	8.39	0.4116 ± 0.0048
64	4	12672	3.900 ± 0.039	10.85	0.3938 ± 0.0044
64	6	20992	4.058 ± 0.043	15.67	0.3882 ± 0.0044
64	8	29312	4.368 ± 0.049	20.80	0.3643 ± 0.0042
64	12	45952	4.865 ± 0.062	28.88	0.3431 ± 0.0040
64	16	62592	5.100 ± 0.075	36.46	0.3433 ± 0.0046

- Theorem 1: `Sandwich`, `nearestDist_le_verticalDist`, `verticalDist_le_of_lipschitz`.
- Theorem 2 and Corollary 3: `Marginal`, `expected_residualLoss_congr`; `RandomPhase`, `marginal_randomPhaseWave_eq`; `Verdict`, `not_learningClaim_sineWave`.
- Theorem 5: `Coverage`, `integral_infDist_eq_integral_uncovered`.
- Theorem 11: `Oscillation`, `nearestDist_le_of_coversWithin`, `verticalDist_le_of_lipschitz_of_coversWithin`, `expected_nearestDist_le`.
- Theorem 12: `SineCloser`, `sineWave_nearestDist_bounds`, `tendsto_expected_nearestDist_sineWave`; collected in `SquigglyLineTheorem`, `squiggly_line_theorem`.
- Propositions 13 and 16: `LinearModel`, `residual_along`, `exists_within_budget`, `travel_coprod_le`.
- Theorem 14: `Kernel`, `residual_gradientStep`, `travel_eq_norm_of_kernelInverse`, `travel_eq_of_kernel_eq`, `residual_iterate_gradientStep_eq`.
- Corollary 15: `ExchangeRate`, `TightFeatures.travel_eq`; `MarkovChains`, `TightFeatures.residual_iterate_gradientStep`, `TightFeatures.residual_gradientStep_inv`.
- Appendix B: `NarrowChain`, `narrowChain_neverTurns`; `Depth`, `iterate_tent_dyadic`; `Bridge`, `sin_sq_iterate_tent`; `Polarizer`, `relu2_nearest`, `fan_of_le_pi_div_two`, `fan_of_pi_div_two_le`, `iterate_fan`; `ThreePolarizers`, `malus`, `throughPlates_light`, `three_polarizers`, `quarter_turn_arrives`; `MarkovChains`, `merged_stays_merged`, `card_image_runLayers_append_le`, `dead_stays_dead`.